

Project: Digital Wallet Transaction Analysis

1. Dataset Overview

Dataset name: Digital Wallet Transactions

Source: Kaggle (<https://www.kaggle.com/datasets/harunrai/digital-wallet-transactions/>)

This dataset is a synthetic dataset for a virtual digital wallet company. It simulates 5,000 transaction records with 16 attributes associated to each record.

The 5,000 transaction records occurred from 2023-08-19 to 2024-08-18, one year timeframe.

The 16 attributes (columns) are:

1. idx
2. transaction_id
3. user_id
4. transaction_date
5. product_category
6. product_name
7. merchant_name
8. product_amount
9. transaction_fee
10. cashback
11. loyalty_points
12. payment_method
13. transaction_status
14. merchant_id
15. device_type
16. location

With this dataset, I aim to discover:

1. Which product category has the most transaction fee income?
2. Which product category has the most fee yield ?
3. What device type has the least transaction successful rate?

With the analysis result, I can then make suggestions to the company to determine strategies that help to increase the company's revenue.

2. Import dataset

Import "digital_wallet_transactions" dataset to BigQuery (My database program).

3. Data quality check

This dataset is ready for analysis when it was uploaded to Kaggle, so means it has been cleaned. Yet some simple data quality checks were conducted just to confirm its readiness.

1. Review data type of all columns

- After reviewing the data type of all columns, everything is correct.

Schema	Details	Preview	Table Explor
☐	Field name	Type	
☐	idx	INTEGER	
☐	transaction_id	STRING	
☐	user_id	STRING	
☐	transaction_date	TIMESTAMP	
☐	product_category	STRING	
☐	product_name	STRING	
☐	merchant_name	STRING	
☐	product_amount	FLOAT	
☐	transaction_fee	FLOAT	
☐	cashback	FLOAT	
☐	loyalty_points	INTEGER	
☐	payment_method	STRING	
☐	transaction_status	STRING	
☐	merchant_id	STRING	
☐	device_type	STRING	
☐	location	STRING	

2. To make sure the 5,000 transaction IDs are unique

- I used the DISTINCT keyword in SQL and verified that the transaction IDs are all unique, meaning that the dataset really recorded 5,000 different transactions.

Untitled query

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```
1 SELECT
2   COUNT (DISTINCT transaction_id)
3 FROM
4   `project1-582206.digital_wallet.digital_wallet_transactions`
```

Query completed

Using on-demand processing quota

Query results Chat View all jobs Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	f0_
1	5000

3. To check is there any negative value in numeric columns, there are four columns should be checked which are “product_amount”, “transaction_fee”, “cashback” and ”loyalty_points”.

- The four columns do not contain negative value.

Untitled query Run Save Schedule Share

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```

1 SELECT
2   COUNTIF(product_amount < 0),
3   COUNTIF(transaction_fee < 0),
4   COUNTIF(cashback < 0),
5   COUNTIF(loyalty_points < 0)
6 FROM
7   `project1-502206.digital_wallet.digital_wallet_transactions`

```

Query completed

Using on-demand processing quota

Query results Chat View all jobs Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	f0_	f1_	f2_	f3_
1	0	0	0	0

4. To identify top product category that generates most transaction fees

To use SQL to extract transaction fees data from each product category, and then sort them in descending order.

- After running SQL query, Streaming Service has contributed the most transaction fees, \$7753.90, made up 6.16% of total transaction fees.

Untitled query Run Save Schedule Share

File Edit Run Tools

```

1 SELECT
2   product_category,
3   ROUND (
4     SUM(transaction_fee),
5     2
6   )
7   AS fee_income,
8   ROUND (
9     SUM (transaction_fee) / SUM(SUM(transaction_fee)) OVER() * 100,
10    2
11   )
12   AS fee_income_percentage
13 FROM
14   `project1-502206.digital_wallet.digital_wallet_transactions`
15 GROUP BY
16   product_category
17 ORDER BY
18   fee_income DESC

```

Query completed

Using on-demand processing quota

Query results Chat View all jobs Save results Open in

Job information Results Visualization JSON Execution details Execution graph

Row	product_category	fee_income	fee_income_perc...
1	Streaming Service	7753.9	6.16
2	Education Fee	7449.83	5.92
3	Hotel Booking	6876.42	5.46
4	Taxi Fare	6622.4	5.26

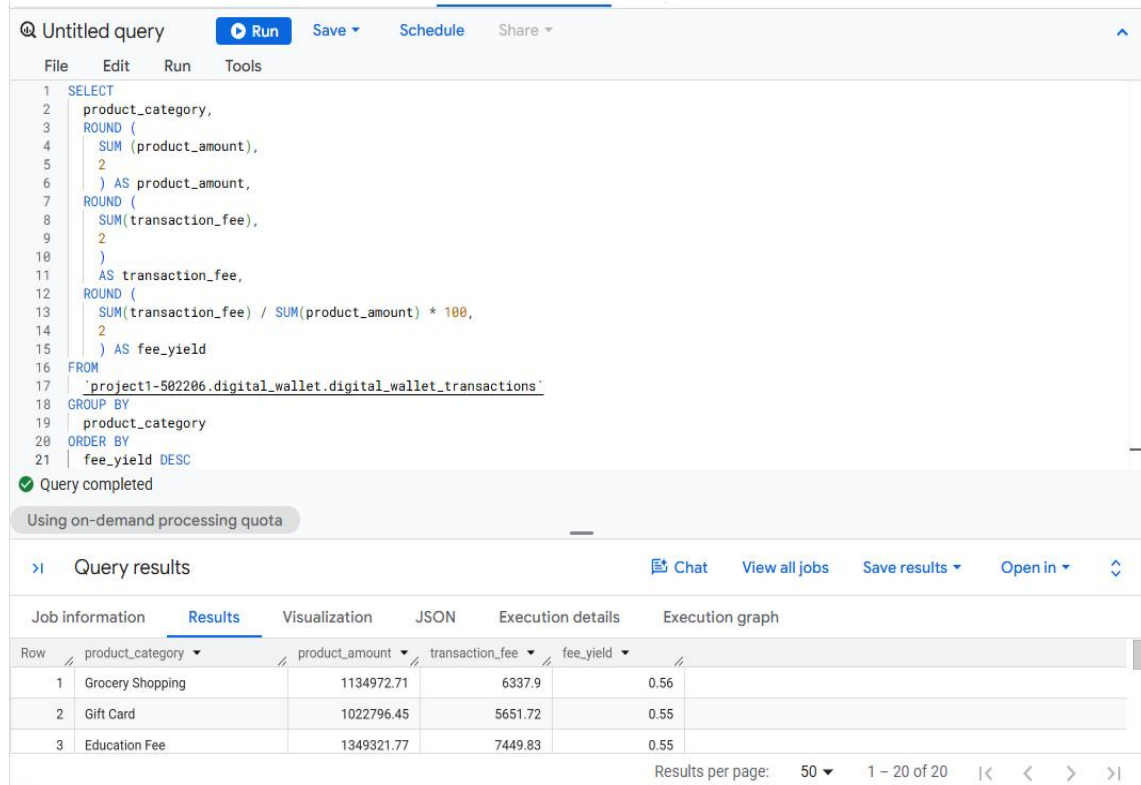
Results per page: 50 1 – 20 of 20

5. To identify which product category has the highest fee yield

Fee yield refers to how many percent the company earns from product amount (GMV).

To use SQL to extract transaction fees and product amount from each product category. Then, divide the transaction fee by product amount, multiply by 100, to get the fee yield, and then sort them in descending order.

- Grocery Shopping has the highest fee yield, 0.56%. Meaning that on average, the company charge 0.56% for transaction on Grocery Shopping.



The screenshot shows a SQL query editor with a query to calculate the fee yield for different product categories. The query is as follows:

```
1 SELECT
2   product_category,
3   ROUND (
4     SUM (product_amount),
5     2
6   ) AS product_amount,
7   ROUND (
8     SUM(transaction_fee),
9     2
10  )
11  AS transaction_fee,
12  ROUND (
13    SUM(transaction_fee) / SUM(product_amount) * 100,
14    2
15  ) AS fee_yield
16 FROM
17   'project1-502206.digital_wallet.digital_wallet_transactions'
18 GROUP BY
19   product_category
20 ORDER BY
21   fee_yield DESC
```

Below the query editor, a status bar indicates "Query completed" and "Using on-demand processing quota".

The results are displayed in a table with the following columns: Row, product_category, product_amount, transaction_fee, and fee_yield. The table shows three rows of data:

Row	product_category	product_amount	transaction_fee	fee_yield
1	Grocery Shopping	1134972.71	6337.9	0.56
2	Gift Card	1022796.45	5651.72	0.55
3	Education Fee	1349321.77	7449.83	0.55

At the bottom right of the results table, it says "Results per page: 50" and "1 - 20 of 20".

6. To identify which device type has the least transaction successful rate

There are 3 types of device: Android, iOS, Web

And 3 types of transaction status: Successful, Failed, Pending

To use SQL to extract each device type with number of successful, failed and pending transactions. Then, calculate the transaction status rate.

- The result tells us that transactions conducted by Web-based platform has the least successful rate which is 93.74%, and 4.18% failed, 2.09% pending.

Untitled query				
Run Save Schedule Share				
FileEditRunTools				
<pre> 1 SELECT 2 device_type, 3 transaction_status, 4 COUNT (transaction_status) AS quantity, 5 ROUND(6 COUNT (transaction_status) / SUM(COUNT(transaction_status)) OVER(PARTITION BY device_type) * 100, 7 2 8) AS transaction_status_rate 9 FROM 10 `project1-502206.digital_wallet.digital_wallet_transactions` 11 GROUP BY 12 Device_type, transaction_status </pre>				
This query will process 93.85 KB when run. Using on-demand processing quota				
Query results ChatView all jobsSave resultsOpen in				
Job informationResultsVisualizationJSONExecution detailsExecution graph				
Row	device_type	transaction_status	quantity	transaction_statu...
1	Android	Successful	2867	95.73
2	Android	Failed	72	2.4
3	Android	Pending	56	1.87
4	Web	Successful	449	93.74
5	Web	Pending	10	2.09
6	Web	Failed	20	4.18
7	iOS	Successful	1439	94.3
8	iOS	Pending	33	2.16
9	iOS	Failed	54	3.54
Results per page: 50 1 - 9 of 9				

7. Business Recommendation

As a conclusion, we know that transactions for Streaming Service is currently the largest source of transaction fee income for the company. If the company wants to optimize this source of income, the company may build an “auto pay” function in the digital wallet platform for users to make periodic payment to Streaming Service merchants automatically, to strengthen this largest source of income.

Furthermore, Grocery Shopping has the highest fee yield for the company. It is worth investigating why Grocery Shopping merchants are willing to bear higher cost for digital wallet payment solutions. And then, the company might gain some insights to increase fee yields on all product categories.

Finally, the transaction successful rate on Web-based payments is the least. It would be great if IT teams can manage to optimize the Web-based payment process, to safeguard the revenue that the company supposed to earn.